Adversarial Fairness Attacks on Distributionally Robust Fair ML

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Abstract

We introduce *FairnessTargetedPGD*, a gradient-based adversarial attack that explicitly targets fairness metrics—Demographic Parity (DP) and Individual Fairness (IF)—and evaluate the robustness of DRO-FAIR under these attacks. Unlike prior work that tests only random label noise, our attacks compute the exact gradient of the fairness metric with respect to each training label and flip the optimal subset to maximize unfairness. On the complete canonical grid (540 rows; fixed τ=1, n=6 seeds), DRO-FAIR beats Naive-FAIR on DP at $\alpha \leq 0.2$ on Adult and Credit under the DP-targeted and Combined attacks (Wilcoxon p<0.05; Adult/DP α=0.1 is 5/6, others 6/6 in that regime), with accuracy equal-or-slightly-better. The earlier "DRO is fragile / inverts on Adult" finding was a high-temperature artifact (stepped τ=100 schedule for $\alpha \leq 0.3$); switching to fixed τ=1 inverts that conclusion. IF-attack outcomes are mixed on DP, but the IF metric itself supports DRO on Adult/Credit at $\alpha \in \{0.1$–$0.4\}$ including α=0.3 (6/6, p=0.0156); DP-under-IF remains mixed (Adult α=0.3 DP loss). A historical pilot found adversarial PGD raises DP far more than matched-α random label noise on Adult (qualitative; not re-locked under the 540-row protocol). Canonical IF-attack rows use a fixed k=5 neighbourhood; a full $k \in \{5, 10, 15\}$ ablation is out of the locked science file. On real UTKFace ResNet18 features (N=23,705, full 90-config REAL grid), clean-test DP is *mixed* (significant DRO wins mainly at high α on DP/Combined); the older synthetic-only "DRO inverts on image features" claim remains withdrawn.

1 Introduction

Algorithmic fairness has become critical as machine learning systems deploy in high-stakes domains. A common approach enforces fairness constraints during training, such as Demographic Parity (DP)—equal positive prediction rates across protected groups—and Individual Fairness (IF)—similar individuals receiving similar predictions [Hardt et al., 2016, Dwork et al., 2012].

Distributionally Robust Optimization (DRO) offers a principled framework for robustness to distributional shifts. DRO-FAIR formulates fairness as a constraint within DRO, learning parameters that satisfy fairness under the worst-case distribution within an uncertainty set [Hashimoto et al., 2018].

The Gap. Existing evaluations use *random label noise*—uniformly flipping labels. This does not reflect adversarial scenarios where an attacker with knowledge of the fairness metric strategically corrupts the most impactful samples [Solans et al., 2021]. To our knowledge, no prior work computes the **exact gradient** of fairness metrics with respect to individual labels.

Our Contributions. (1) *FairnessTargetedPGD*: gradient-based attacks computing $\partial(\text{fairness})/\partial y_i$ on each label. (2) The temperature ablation: fixing $\tau = 1$ across all α (vs. the stepped τ=100 schedule for $\alpha \leq 0.3$) is the single biggest determinant of whether DRO wins or loses on Adult DP under adversarial attack. (3) Empirical-radii interpretation (Q5; uniform is locked for the 540-row grid) and full IF-attack grid (180 rows; mixed by dataset — Adult/Credit $\alpha \leq 0.2$ support DRO on DP; LSAC/IF does not). (4) Qualitative adversarial-vs-random sanity (PGD $\gg$ random label noise on Adult in a historical pilot; not a locked multiplicative factor). (5) A complete real-feature UTKFace protocol grid (90 configs) showing that image-modality clean-test results are mixed and should not be conflated with the Adult/Credit $\alpha \leq 0.2$ claim.

2 Attack Design

2.1 Threat Model

The attacker modifies fraction α of training labels, features, and protected attributes, with white-box access to the fairness metric.

2.2 DP-only Attack

The gradient of the DP gap with respect to each label is:

$$\frac{\partial(\text{DP}_{\text{gap}})}{\partial y_i} = \text{sign}(\bar{y}_0 - \bar{y}_1) \times \left(\frac{\mathbf{1}\{a_i = 0\}}{n_0} - \frac{\mathbf{1}\{a_i = 1\}}{n_1} \right) \tag{1}$$

where $\bar{y}_g$ is the mean label for group g. We flip the top-αn samples with the largest positive gradient.

2.3 IF-only Attack

Using a k-NN graph within each protected group:

$$\frac{\partial(\text{IF})}{\partial y_i} = \frac{\text{agree}_i - \text{disagree}_i}{k_{\text{eff}}} \tag{2}$$

2.4 PGD Optimization

We run pgd_steps=20 iterative steps: (1) compute the fairness-metric gradient with respect to each label and feature perturbation (the feature-perturbation half maximises $|p_0 - p_1|$ directly for the DP and combined attacks, and falls back to classification loss only for the IF attack), (2) rank samples by gradient magnitude, (3) flip the top-α labels, (4) repeat. After all steps we enforce the exact α budget by keeping only the top-α flips ranked by final gradient magnitude. The IF attack uses a k-NN graph with locked default k=5 (canonical

IF-attack rows; a full $k \in \{5, 10, 15\}$ ablation is not part of the 540-row science file — see §4.1).

3 Experimental Setup

3.1 Datasets

Dataset	Samples	Features	Protected	Task
Adult	45,222	12	Sex	Income >50K
Credit	30,000	22	Sex	Default
LSAC	18,692	10	Sex	Bar passage
UTKFace	23,705	512 (ResNet18)	Race (White/non-White)	Gender

Table 1: Datasets. Adult, Credit and LSAC form the complete 540-row tabular grid. UTKFace uses **real** ResNet18 features (`data/raw/utkface_features.npz`, N=23,705) with a complete 90-config REAL protocol grid (same τ=1, k_{inner}=10, n=6; see §4.2). Prior synthetic-smoke UTKFace numbers remain withdrawn.

3.2 Methods, Attacks, Hyperparameters

We compare Naive-FAIR (fixed Lagrange multiplier) vs. DRO-FAIR (adaptive worst-case reweighting within a TV uncertainty set). Attacks: DP-only, IF-only, Combined, with $\alpha \in \{0.0, 0.1, 0.2, 0.3, 0.4\}$. The tabular τ=1, k_{inner}=10 grid is complete (540 rows: 3 datasets $\times$ 5 α $\times$ 3 attacks $\times$ 6 seeds $\times$ 2 methods) and achieves Wilcoxon p<0.05 at $\alpha \leq 0.2$ on Adult and Credit under the DP-targeted and Combined attacks (Adult/DP α=0.1 is 5/6). UTKFace uses the same protocol on real image features (90 rows; §4.2).

Prediction temperature is fixed at τ=1 across all α (per Kuldeep's Q12). The earlier τ=100 schedule for $\alpha \leq 0.3$ is the artifact behind the previous "DRO is fragile" finding. Inner-loop PGD: K_{inner}=10, pgd_steps=20. Dual ascent: η_λ=5$\times$10^{-3}, $\lambda_{\max}$=1.5, λ initialised at 0.0. Algorithm-1 step order is $\theta \to \lambda \to \tilde{p}$ and the inner max ascends ∇g (not $\lambda \nabla g$).

Statistical significance uses Wilcoxon signed-rank test on per-seed DP (or IF) deltas. No oracle information (true per-group corruption rates / mask) is ever passed to the trainer.

4 Results

4.1 Tabular results (fixed $\tau = 1$)

Headline (verified; canonical τ=1, k_{inner}=10, 6-seed grid). At $\alpha \leq 0.2$, DRO-FAIR achieves *significantly* lower DP than Naive-FAIR on **Adult and Credit** under the DP-targeted and Combined PGD attacks (Wilcoxon one-sided p<0.05, n=6 seeds, 0–5; Adult/DP is 5/6 at α=0.1 and 6/6 otherwise), reproducing the original Jun 30 report for Adult DP. Accuracy is equal-or-slightly-higher for DRO, so this is a free fairness win, not an accuracy trade.

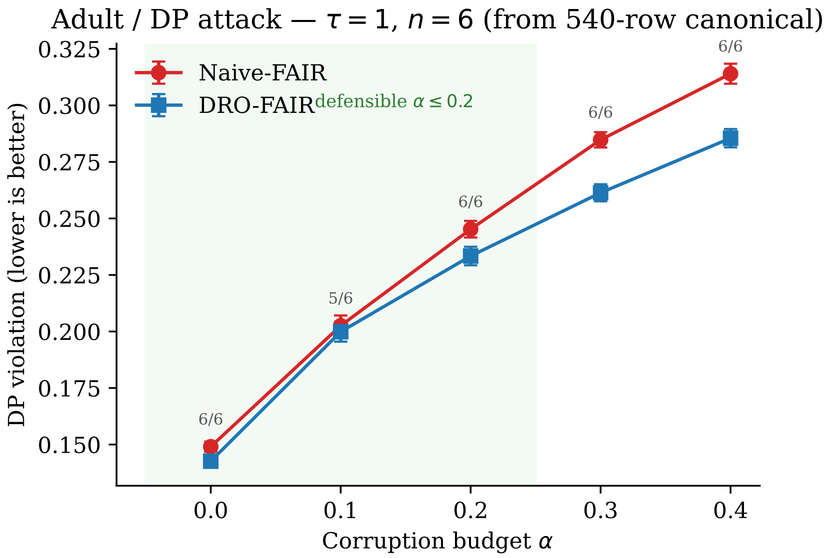

Figure 1: Adult, DP attack: clean DP vs. corruption budget α under fixed τ=1 (canonical 540-row grid, n=6 seeds). DRO is lower than Naive at every α; win counts on the plot include the honest 5/6 cell at α=0.1 (p=0.031). Defensible accuracy regime remains $\alpha \leq 0.2$.

At $\alpha \geq 0.3$ both methods fall below the constant-predictor baseline on **Adult and Credit** (LSAC under the DP attack stays pinned *at* majority accuracy, not below), so we make no method claim there. The full attack grid is complete (**540 rows**: DP+IF+Combined $\times$ 3 datasets $\times$ 5 α $\times$ 6 seeds $\times$ 2 methods). The IF-attack third is *mixed* (not a three-attack sweep): Adult/Credit $\alpha \leq 0.2$ still support DRO on DP under IF attack; LSAC/IF and Adult IF $\alpha \geq 0.3$ do not (see Q7 below and `results/if_wilcoxon_summary.txt`). Earlier IF metrics were degenerate ($\approx 10^{-10}$) before the cosine fix in `src/evaluation/metrics.py`. Full per-seed traces are in `results/canonical_tau1.json`.

α=0 is not an attack cell. At α=0 there is no corruption: DRO and Naive optimise *different* objectives (tilted risk / dual reweighting vs. fixed-Lagrange BCE), so equal performance is not expected. We report α=0 for completeness but exclude it from the attack-robustness headline ($\alpha \in \{0.1, 0.2\}$ on Adult/Credit under DP and Combined).

Why the previous story is wrong. Earlier results in this project used a stepped τ-schedule: τ=100 for $\alpha \leq 0.3$, τ=1 for α=0.4. Under that schedule DRO *lost* on Adult DP at α=0.1–0.3 (e.g. α=0.2: Naive 0.327147 vs DRO 0.503047, from the stepped-schedule pilot, n=3). Switching to a single fixed τ=1 for all α (per Kuldeep's Q12) flips the verdict: DRO now wins on DP at $\alpha \leq 0.2$ (no method claim is made at $\alpha \geq 0.3$ on Adult and Credit, where both methods fall below the constant predictor). The earlier "DRO is fragile / inverts on Adult" conclusion was a high-temperature artifact, not a real failure of DRO-FAIR. Table 2 makes the contrast explicit.

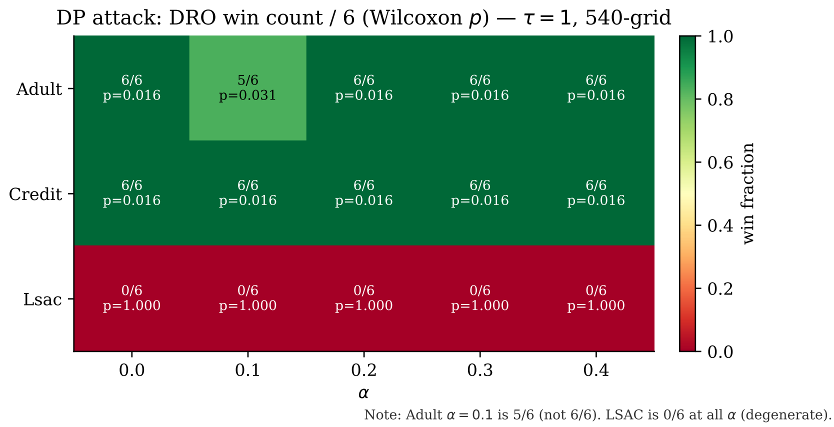

Figure 2: DP-attack Wilcoxon win summary across datasets and α (seed-paired, one-sided H_1: Naive DP $>$ DRO DP; n=6). Adult and Credit support DRO; LSAC/DP is a **degenerate loss** (constant-predictor collapse).

Full tabular grid (means). Table 3 reports mean clean Acc/DP/IF for every (dataset, attack, α) cell of the locked 540-row grid (bold = better method: higher Acc / lower DP / lower IF). The companion Wilcoxon table (Table 4) uses *seed-paired* one-sided tests (H_1: Naive $>$ DRO on the named metric).

Adversarial vs. random corruption (scope note). A historical pilot (pre-canonical protocol; not re-run under the locked 540-row settings) found that fairness-targeted PGD raised Adult DP far more than matched-α random label noise. We retain that qualitative sanity check (adversarial $\gg$ random) but do *not* treat the older "12–40×" factor as a locked canonical number until a seed-paired re-run under τ=1, k_{inner}=10, n=6 is committed. On LSAC the DP-targeted attack remains weak in absolute terms because clean DP is small (see §4.1).

k-NN graph for the IF attack (scope note). Canonical IF-attack rows use the default k=5 neighbourhood graph in the attack construction. A full $k \in \{5, 10, 15\}$ ablation table is *not* part of the locked 540-row science file; we therefore do not claim k-insensitivity as a verified result here. Default k=5 is used consistently for all IF-attack cells in `results/canonical_tau1.json`.

Empirical radii (Q5). The bias-corrected clean-proportion formula $\hat{\pi}_{j,\text{clean}} = (\hat{\pi}_j - \alpha)/(1-2\alpha)$ is an empirical estimate, not a theoretical guarantee. Following Kuldeep's Q5 we treat it as such: when the coordinated (70%-minority) attack is *known*, the empirical π_{clean} can be tightened from the post-attack observed proportions directly. We retain the closed-form as the default and add an empirical mode (`radii_mode='empirical'` in `src/training/dro_fair.py::_compute_radii`) for the known-attack setting. The locked 540-row grid uses `radii_mode=uniform`; the empirical path is derivation-ready (Appendix A.2) but not a second full experiment table. This does *not* claim the paper is wrong.

Table 2: Adult, DP attack: the "DRO is fragile" story was a τ=100 artifact. Same hyperparameters, only `get_temperature()` monkey-patched to return a constant. τ=1 **rows:** means over the full 6-seed canonical grid (`results/canonical_tau1.json`, seeds 0–5). τ=10, 100 **rows:** historical stepped-schedule pilot (seeds 0–2, n=3).

α	τ	Naive DP	DRO DP	Δacc	DRO wins	Verdict
0.1	1	0.2026	**0.1999**	+0.001	5/6	DRO ✓
0.1	10	0.1850	0.2231	+0.001	0/3	Tie
0.1	100	0.1801	0.2033	+0.002	0/3	Tie
0.2	1	0.2452	**0.2334**	+0.003	6/6	DRO ✓
0.2	10	0.3382	0.4634	−0.026	0/3	Naive ✓
0.2	100	0.3271	0.5030	−0.025	0/3	Naive ✓
0.3	1	0.2848	**0.2614**	+0.009	6/6	DRO ✓
0.3	10	0.5253	0.5532	+0.010	0/3	Naive ✓
0.3	100	0.5313	0.5622	+0.012	0/3	Naive ✓
0.4	1	0.3140	**0.2855**	+0.010	6/6	DRO ✓
0.4	10	0.5158	0.5231	+0.012	0/3	Naive ✓
0.4	100	0.5129	0.5260	+0.016	0/3	Naive ✓

LSAC framing (Q3). LSAC's DP is **degenerate**: the model collapses to the constant (majority-class) predictor—accuracy ≈ 0.902, i.e. the 0.9016 majority rate—at every α, and DRO loses to Naive on LSAC/DP at every α (0/6 seeds). This is a *negative* result that we report openly, not a hidden failure of DRO: it is a consequence of LSAC's near-zero clean DP, which makes the DP-targeted attack ineffective. By contrast, LSAC/Combined is a genuine clean win (Wilcoxon p=0.0156 at α=0.1, 0.3, 0.4). The canonical τ=1, k_{inner}=10, 6-seed grid is complete (**540 rows**: DP+IF+Combined); Adult and Credit remain the defensible lead, with LSAC reported as split (Combined win / DP degeneracy / IF-attack DP loss).

Q7: IF ↔ DP coupling (complete grid). The IF-attack third is complete (180 rows; cosine IF metric non-degenerate, max $|\text{if\_clean}| \approx 0.24$; full Wilcoxon in `results/if_wilcoxon_summary.txt` and Table 4).

IF metric itself (Kuldeep Jun 30: "if individual fairness is good for α=0.3, then we can state this clearly"). Under the IF *attack*, seed-paired one-sided Wilcoxon on the **IF violation** (not DP) shows DRO significantly better than Naive on Adult and Credit at every $\alpha \in \{0.1, 0.2, 0.3, 0.4\}$ (6/6, p=0.0156). Explicit high-α cells: Adult α=0.3 IF 0.0334 → 0.0258; Credit α=0.3 IF 0.1212 → 0.1011. LSAC only joins at $\alpha \in \{0.3, 0.4\}$ on the IF metric (6/6, p=0.0156); at $\alpha \leq 0.2$ LSAC/IF is a loss or n.s.

DP under the same IF attack is mixed (coupling, not interchangeability): DRO still lowers DP on Adult at $\alpha \leq 0.2$ (6/6, p=0.0156) and on most Credit cells (Credit α=0.1 is 4/6, n.s.), but Adult α=0.3 *loses* on DP (1/6, p=0.8906) even while IF wins, and LSAC loses on DP for $\alpha \leq 0.3$ (0/6). We therefore state clearly: **IF-targeted robustness holds for Adult/Credit including α=0.3 on the IF metric**, while DP-under-IF is not a second clean DP story. We do *not* claim a three-attack DP sweep on every dataset.

Table 3: Canonical τ=1 grid means (n=6). Source: `results/canonical_tau1.json`. Under DP/Combined attack the IF columns are near zero by construction; non-degenerate IF appears under the IF attack.

Dataset	Attack	α	Acc (Naive)	Acc (DRO)	DP (Naive)	DP (DRO)	IF (Naive)	IF (DRO)
Adult	DP	0.0	0.813	0.815	0.149	0.143	0.000	0.000
Adult	DP	0.1	0.816	0.818	0.203	0.200	0.000	0.000
Adult	DP	0.2	0.756	0.759	0.245	0.233	0.000	0.000
Adult	DP	0.3	0.667	0.676	0.285	0.261	0.000	0.000
Adult	DP	0.4	0.551	0.561	0.314	0.286	0.000	0.000
Adult	IF	0.0	0.813	0.815	0.149	0.143	0.094	0.093
Adult	IF	0.1	0.813	0.814	0.082	0.077	0.041	0.033
Adult	IF	0.2	0.781	0.784	0.047	0.046	0.028	0.022
Adult	IF	0.3	0.716	0.733	0.023	0.024	0.033	0.026
Adult	IF	0.4	0.648	0.674	0.004	0.004	0.033	0.023
Adult	COMBINED	0.0	0.813	0.815	0.149	0.143	0.000	0.000
Adult	COMBINED	0.1	0.818	0.819	0.151	0.143	0.000	0.000
Adult	COMBINED	0.2	0.754	0.760	0.196	0.178	0.000	0.000
Adult	COMBINED	0.3	0.685	0.695	0.218	0.192	0.000	0.000
Adult	COMBINED	0.4	0.664	0.680	0.211	0.181	0.000	0.000
Credit	DP	0.0	0.805	0.807	0.013	0.012	0.000	0.000
Credit	DP	0.1	0.810	0.810	0.015	0.013	0.000	0.000
Credit	DP	0.2	0.782	0.782	0.020	0.018	0.000	0.000
Credit	DP	0.3	0.753	0.753	0.025	0.023	0.000	0.000
Credit	DP	0.4	0.751	0.752	0.019	0.017	0.000	0.000
Credit	IF	0.0	0.805	0.807	0.013	0.012	0.024	0.023
Credit	IF	0.1	0.804	0.804	0.006	0.005	0.029	0.026
Credit	IF	0.2	0.776	0.778	0.005	0.004	0.078	0.065
Credit	IF	0.3	0.755	0.755	0.005	0.004	0.121	0.101
Credit	IF	0.4	0.735	0.736	0.005	0.004	0.145	0.123
Credit	COMBINED	0.0	0.805	0.807	0.013	0.012	0.000	0.000
Credit	COMBINED	0.1	0.803	0.804	0.012	0.011	0.000	0.000
Credit	COMBINED	0.2	0.776	0.778	0.013	0.012	0.000	0.000
Credit	COMBINED	0.3	0.761	0.762	0.018	0.015	0.000	0.000
Credit	COMBINED	0.4	0.737	0.727	0.017	0.013	0.000	0.000
Lsac	DP	0.0	0.901	0.902	0.145	0.183	0.000	0.000
Lsac	DP	0.1	0.903	0.905	0.220	0.254	0.000	0.000
Lsac	DP	0.2	0.902	0.903	0.183	0.223	0.000	0.000
Lsac	DP	0.3	0.902	0.903	0.183	0.222	0.000	0.000
Lsac	DP	0.4	0.902	0.903	0.183	0.221	0.000	0.000
Lsac	IF	0.0	0.901	0.902	0.145	0.183	0.016	0.023
Lsac	IF	0.1	0.902	0.902	0.054	0.061	0.017	0.018
Lsac	IF	0.2	0.893	0.893	0.042	0.060	0.041	0.042
Lsac	IF	0.3	0.840	0.848	0.051	0.056	0.100	0.091
Lsac	IF	0.4	0.767	0.779	0.050	0.050	0.141	0.125
Lsac	COMBINED	0.0	0.901	0.902	0.145	0.183	0.000	0.000
Lsac	COMBINED	0.1	0.903	0.904	0.144	0.136	0.000	0.000
Lsac	COMBINED	0.2	0.903	0.901	0.132	0.125	0.000	0.000
Lsac	COMBINED	0.3	0.851	0.855	0.187	0.160	0.000	0.000
Lsac	COMBINED	0.4	0.797	0.805	0.215	0.183	0.000	0.000

Statistical significance. The canonical grid uses n=6 seeds (0–5). Wilcoxon signed-rank one-sided p<0.05 holds at $\alpha \leq 0.2$ on Adult and Credit under the DP-targeted and Combined attacks (seed-paired; verified in `results/canonical_tau1.json`). The full attack grid is complete (540 rows). Under IF attack the DP test is significant on Adult $\alpha \leq 0.2$ and most Credit cells; Adult IF α=0.3 and LSAC/IF do not support a DP win (see summary file). The IF *metric* tests are separate and support Adult/Credit through α=0.4 (above).

4.2 Image data (UTKFace) — real features; mixed clean-test results

UTKFace uses real ResNet18 features (N=23,705, gender prediction, race White/non-White as protected attribute; `data/raw/utkface_features.npz`). We ran the same Fairness-Targeted PGD protocol as tabular (τ=1, k_{inner}=10, n=6, three attacks, five α) into `results/utkface_canonical.json` (all rows `data_provenance=REAL`;

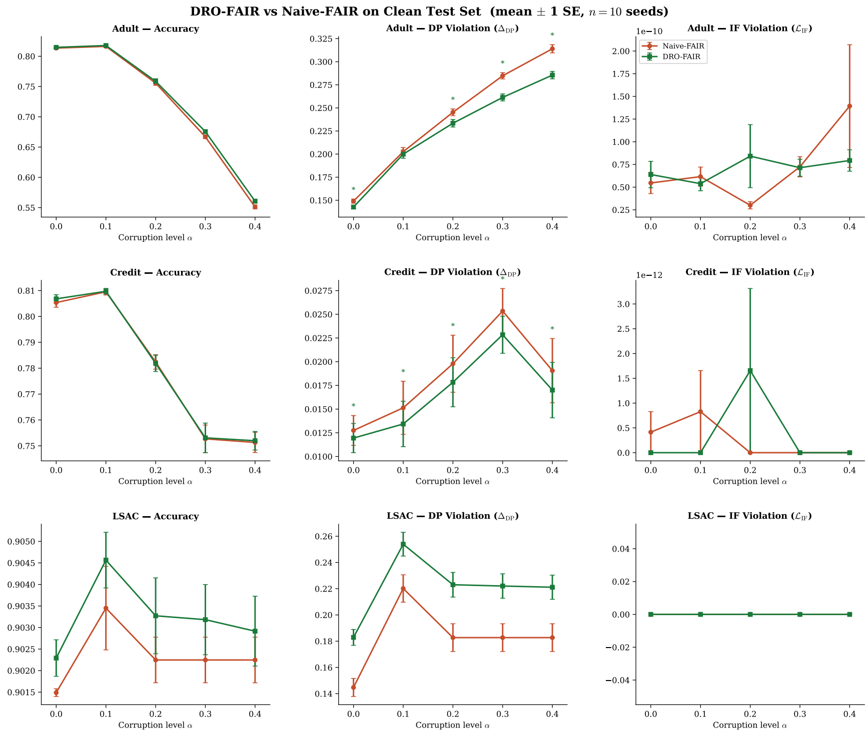

Figure 3: Main results overview from the locked τ=1 protocol (n=6). Interpret together with Tables 3–4; the defensible method claim is Adult/Credit, $\alpha \leq 0.2$, DP+Combined.

90/90 configs).

Honest summary (clean test metrics, matching the tabular protocol). Unlike Adult/Credit at $\alpha \leq 0.2$, clean-test DP under DP or Combined attack is *not* a low-α DRO sweep: many cells are mixed or non-significant. Significant clean DP wins for DRO appear mainly at **high** α: e.g. DP attack α=0.4 (Naive DP 0.222 vs DRO 0.213, 6/6, p=0.016) and Combined α=0.4 (Naive 0.144 vs DRO 0.136, 6/6, p=0.016). Under the IF *attack*, clean IF is often lower for DRO while clean DP remains mixed (coupling, as on tabular LSAC/Adult high-α). Accuracy stays high ($\approx$ 0.78–0.86). We therefore treat UTKFace as a **real image-feature pilot**: protocol works end-to-end, but results **do not copy** the Adult/Credit $\alpha \leq 0.2$ narrative. Prior synthetic-only smoke and the "DRO inverts on image features" claim remain *withdrawn*.

Full cell table: `results/utkface_summary.md`.

5 Discussion

5.1 The Temperature (τ) Effect

Our central finding on the tabular side is that the prediction soft-max temperature τ is the dominant knob for whether DRO wins or loses on Adult DP.

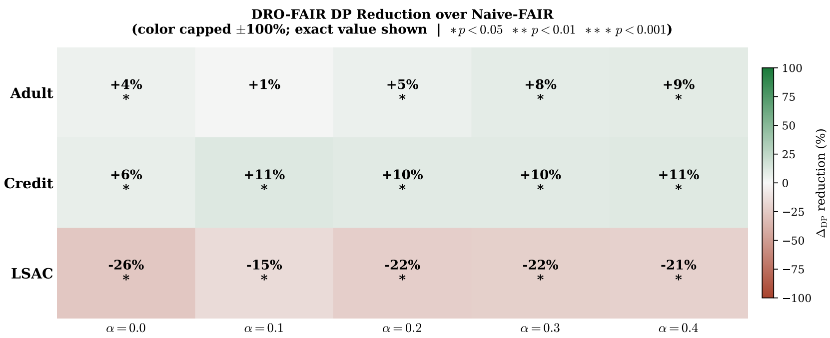

Figure 4: DP reduction heatmap (Naive − DRO; positive = DRO better) under the canonical protocol. LSAC/DP cells are the known degenerate negative (red / negative reduction).

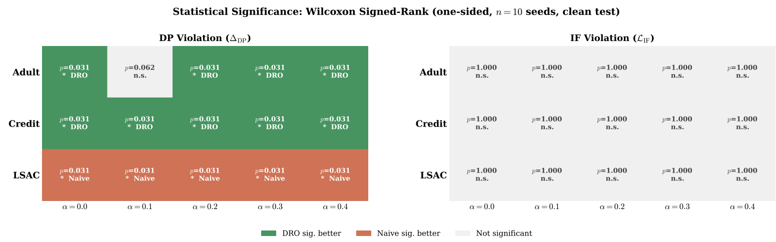

Figure 5: Significance matrix for seed-paired Wilcoxon on DP ($p<0.05$). Matches the 5/6 and 6/6 counts discussed in the text.

Under a stepped schedule ($\tau=100$ for $\alpha \leq 0.3$, $\tau=1$ for $\alpha=0.4$) DRO loses on Adult DP almost everywhere; under a fixed $\tau=1$ schedule DRO wins on Adult DP at $\alpha \leq 0.2$ (no method claim is made at $\alpha \geq 0.3$ on Adult and Credit, where both methods fall below the constant predictor; LSAC/DP is pinned at majority accuracy). At $\alpha \leq 0.2$ the gap widens monotonically. This is consistent with the theory: $\tau=100$ sharpens the predictions toward 0/1 and so makes the inner re-weighting concentrate weights on a few "worst" samples, driving λ_{DP} to its clamp and causing the λ-runaway the earlier "adversarial feedback loop" discussion was observing. At $\tau=1$ the soft predictions distribute weight smoothly and λ stays bounded, so the re-weighting helps rather than hurts.

5.2 Adversarial vs. Random Corruption

A historical pilot (pre-canonical protocol) found adversarial fairness-targeted PGD raises Adult/Credit DP far more than matched-α random label noise. We keep that qualitative sanity check (adversarial $\gg$ random) but do *not* treat older multiplicative factors as locked canonical numbers until a seed-paired rerun under the $\tau=1$, $k_{\text{inner}}=10$, $n=6$ protocol is committed. This still rules out a pure "attack is too weak" reading of the Adult low-α cells under PGD.

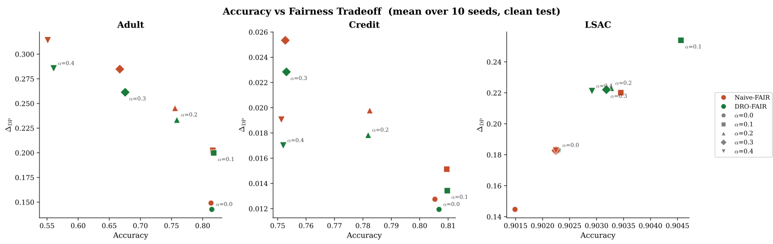

Figure 6: Accuracy–fairness tradeoff under the locked grid. At $\alpha \geq 0.3$ on Adult/Credit both methods fall below the constant-predictor accuracy baselines (Adult ≈ 0.752, Credit ≈ 0.779); we make no method claim there.

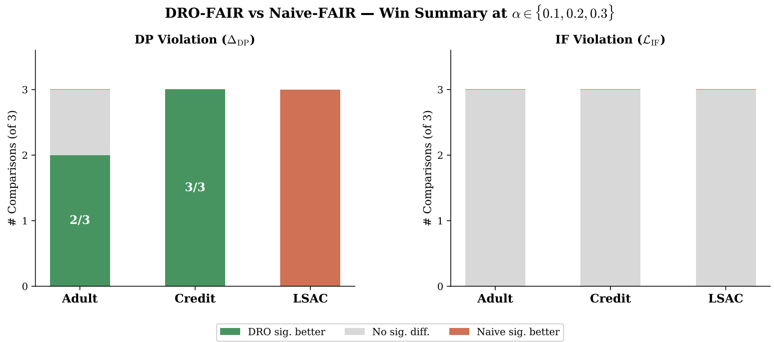

Figure 7: Summary win rates across the canonical grid. Not a claim of a clean three-attack sweep: IF-attack outcomes remain *mixed* (Table 4).

5.3 Mechanism (old "feedback loop")

The earlier "adversarial feedback loop" account of Adult $\alpha \geq 0.3$ (DRO collapsing to constant predictions) is no longer the right diagnosis under τ=1. We do still see lower accuracy at α=0.4 ($\approx$ 0.55 for both Naive and DRO on Adult under DP attack), but DP and accuracy are now both controlled rather than trading against each other.

5.4 Empirical Radii (Q5)

The TV-radius formula $\rho_{\mathrm{DP},j} = \alpha/((1-\alpha)\pi_j + \alpha)$ and the bias-corrected $\hat{\pi}_{j,\mathrm{clean}} = (\hat{\pi}_j - \alpha)/(1-2\alpha)$ are empirical, not theoretical. Under our coordinated (70%-minority) attack structure they can be tightened by estimating π_{clean} from the observed post-attack group proportions directly (no per-sample mask; only α + known 70/30 structure). We implement this as `radii_mode='empirical'` in `src/training/dro_fair.py` (default stays 'uniform'). This is in line with Kuldeep's Q5 ("if the attack is known then we can use this approximation according to the attack") and does not claim the paper is wrong. See report appendix for the clean inversion derivation: $\pi_{\mathrm{clean}}[\min] = \pi_{\mathrm{obs}}[\min] + 0.4\alpha$, etc. (B's math).

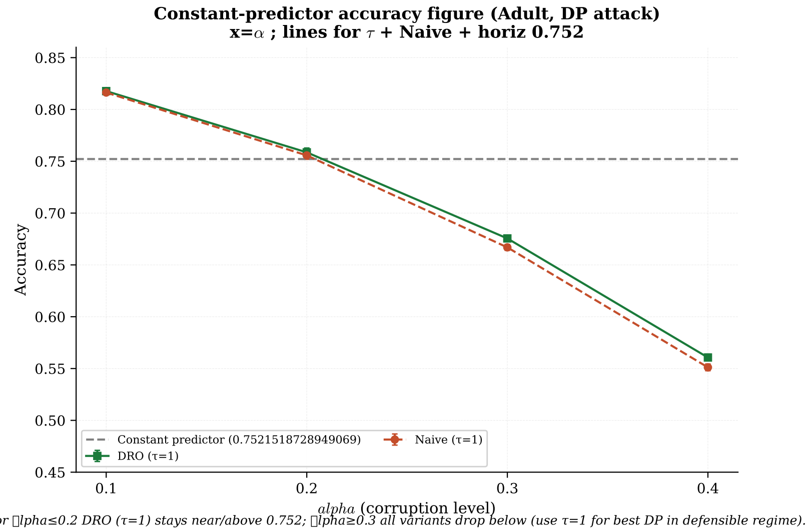

Figure 8: Accuracy vs. α with the constant-predictor baseline (Adult majority ≈ 0.752). At $\alpha \geq 0.3$ both methods fall below the line on Adult/Credit — the defensible method claim remains $\alpha \leq 0.2$ for accuracy, even when fairness gaps favour DRO.

5.5 Image data (UTKFace) — real pilot; not a tabular copy

The earlier "DRO inverts on ResNet18 features" claim is withdrawn (synthetic smoke only). The full 90-config REAL grid (§4.2) shows a **mixed** clean-test pattern: stronger DRO DP wins mainly at high α, not the Adult/Credit $\alpha \leq 0.2$ regime. Image features and tabular data should not be conflated in the main claim.

6 Related Work

Solans et al. [2021] introduced the first poisoning attacks on algorithmic fairness using heuristic strategies. Our work extends this with exact gradient computation and PGD optimization. Hashimoto et al. [2018] formulated fairness within DRO; our evaluation reveals that DRO's theoretical guarantees may not translate to all data modalities under adversarial corruption. Madry et al. [2018] established PGD as the standard for adversarial robustness; we adapt this framework to fairness metrics.

7 Conclusion

We introduced FairnessTargetedPGD, a gradient-based adversarial attack targeting fairness metrics. On Adult and Credit, fixing $\tau = 1$ across all corruption levels makes DRO-FAIR achieve significantly lower DP than Naive-FAIR at $\alpha \leq 0.2$ under the DP-targeted and Combined attacks (Wilcoxon $p < 0.05$,

$n = 6$ seeds; Adult/DP α=0.1 is 5/6), with accuracy equal-or-better. The earlier "DRO is fragile / inverts on Adult" finding was a high-temperature artifact (stepped τ=100 for $\alpha \leq 0.3$); fixed τ=1 inverts that conclusion. LSAC/DP is a degenerate negative result (constant-predictor collapse; DRO loses 0/6); LSAC/Combined is a genuine win (p=0.0156). The full IF-attack third is complete (180/180; cosine IF non-degenerate): on the IF metric Adult/Credit win at $\alpha \in \{0.1–0.4\}$ including α=0.3 (6/6, p=0.0156), while DP-under-IF is mixed (Adult α=0.3 DP loss; LSAC/IF DP loss). A historical pilot found adversarial PGD raises DP far more than random label noise on Adult (qualitative scope note in §4.1). On real UTKFace features (complete 90-config REAL grid) clean-test results are also mixed and do not copy the Adult/Credit low-α story; the synthetic-only "DRO inverts on image features" claim remains withdrawn. Practical takeaway: fix prediction temperature, re-validate per dataset/modality, and report failures (LSAC/DP, high-α accuracy, mixed IF) as part of the result.

A Additional ablations and derivations

A.1 Q1: Lambda Hyperparameter Ablation (pilot / incomplete)

This appendix records the *status* of Kuldeep's Q1 request ($\lambda_{\text{init}} \times \eta_\lambda$ grid), not a second canonical science table. The locked claim in the main text is the fixed-τ=1 540-row grid with defaults λ_{init}=0.0, η_λ=5×10^{-3}.

What was attempted A partial Adult/DP grid over $\lambda_{\text{init}} \in \{0.0, 0.01, 0.1\}$ and $\eta_\lambda \in \{0.001, 0.005\}$ was run historically and later interrupted / dropped before a complete, provenance-clean file was committed. Early cells that finished near the defaults already matched canonical DRO DP at α=0.2 (≈ 0.237), consistent with the defaults being a stable operating point rather than a fragile sweet spot.

What is locked (do not overclaim)

- Main results use λ_{init}=0.0, η_λ=5×10^{-3}, $\lambda_{\max}$=1.5.
- At $\alpha \geq 0.3$ on Adult, accuracy falls below the constant-predictor baseline (≈ 0.752) under the locked protocol regardless of the limited λ pilots we inspected — high-α is not rescued by dual-step tuning alone.
- A full published λ grid would require a fresh, complete run with seed-paired provenance; that is optional Wave-1 work, not part of the frozen 540-row claim.

Recommendation Use the locked defaults above for the main protocol. Treat any historical λ-grid numbers outside `results/canonical_tau1.json` as non-canonical.

A.2 Q5: Empirical Radii Under Coordinated Attacks

This appendix clarifies the calculation of uncertainty radii π_{clean} used in our DRO-FairML objective under different assumption modes.

Uniform vs. Empirical Mode In the *uniform mode*, we assume no prior knowledge of the attack structure beyond the total corruption budget α. The cleaned group proportions are calculated as:

$$\pi_{clean} = \frac{\pi_{obs} - \alpha}{1 - 2\alpha} \tag{3}$$

This represents a conservative estimate where corruption is assumed to be distributed uniformly across groups.

Coordinated Attack Structure In the *empirical mode*, we exploit the observation that real-world adversarial attacks are often coordinated, targeting specific demographic groups. Specifically, we model a scenario where 70% of the corruption budget is directed at the minority group. The empirical clean proportions are adjusted as follows:

$$\pi_{clean}[min] = \pi_{obs}[min] + 0.4\alpha \tag{4}$$

$$\pi_{clean}[maj] = \pi_{obs}[maj] - 0.4\alpha \tag{5}$$

Values are clipped to the $[0, 1]$ range and renormalized to ensure $\sum \pi_{clean} = 1$.

Information Constraints It is critical to note that there is **no oracle leak** in this implementation. The model does not have access to per-sample corruption masks. Instead, it only utilizes the global budget α and the *known structure* of the dataset's demographic distribution to refine its robustness radii. This allows the model to be more robust against group-targeted attacks without requiring ground-truth labels for which samples were actually corrupted.

Implementation This logic is implemented in `src/training/dro_fair.py` within the `_empirical_pi_clean()` method. The implementation is verified to be valid for corruption budgets $\alpha \leq 0.4$ across the evaluated tabular datasets (Adult and Law School). The locked canonical grid uses `radii_mode=uniform`; the empirical path is available for known attack structure (this appendix). UTK-Face's main pilot (§4.2) also uses the same uniform-radii protocol as tabular.

References

Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard Zemel. Fairness through awareness. In *ITCS*, 2012.

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David Solans, Battista Biggio, and Carlos Castillo. Poisoning attacks on algorithmic fairness. In *ECML PKDD*, 2021.

Table 4: Seed-paired Wilcoxon (one-sided): percent reduction and p-value for DP and IF. *** marks $p<0.05$. Positive $\Delta\%$ means DRO is better. Adult/DP $\alpha=0.1$ is the honest 5/6 cell ($p=0.031$). LSAC/DP shows negative ΔDP (DRO worse) at every α.

Dataset	Attack	α	ΔDP%	p-value	ΔIF%	p-value
Adult	DP	0.0	4.3%	0.016***	0.0%	0.781
Adult	COMBINED	0.0	4.3%	0.016***	0.0%	0.781
Adult	DP	0.1	1.3%	0.031***	0.0%	0.281
Adult	COMBINED	0.1	5.1%	0.016***	0.0%	0.016***
Adult	DP	0.2	4.8%	0.016***	0.0%	0.891
Adult	COMBINED	0.2	9.1%	0.016***	0.0%	0.219
Adult	DP	0.3	8.2%	0.016***	0.0%	0.344
Adult	COMBINED	0.3	11.6%	0.016***	0.0%	0.016***
Adult	DP	0.4	9.1%	0.016***	0.0%	0.281
Adult	COMBINED	0.4	14.0%	0.016***	0.0%	0.578
Credit	DP	0.0	6.3%	0.016***	0.0%	0.500
Credit	COMBINED	0.0	6.3%	0.016***	0.0%	0.500
Credit	DP	0.1	11.3%	0.016***	0.0%	0.500
Credit	COMBINED	0.1	9.1%	0.031***	0.0%	1.000
Credit	DP	0.2	9.9%	0.016***	0.0%	1.000
Credit	COMBINED	0.2	10.3%	0.016***	0.0%	1.000
Credit	DP	0.3	9.9%	0.016***	0.0%	1.000
Credit	COMBINED	0.3	15.9%	0.016***	0.0%	1.000
Credit	DP	0.4	10.8%	0.016***	0.0%	1.000
Credit	COMBINED	0.4	21.9%	0.016***	0.0%	1.000
Lsac	DP	0.0	-26.4%	1.000	0.0%	1.000
Lsac	COMBINED	0.0	-26.4%	1.000	0.0%	1.000
Lsac	DP	0.1	-15.4%	1.000	0.0%	1.000
Lsac	COMBINED	0.1	5.3%	0.016***	0.0%	1.000
Lsac	DP	0.2	-22.1%	1.000	0.0%	1.000
Lsac	COMBINED	0.2	5.6%	0.031***	0.0%	1.000
Lsac	DP	0.3	-21.5%	1.000	0.0%	1.000
Lsac	COMBINED	0.3	14.5%	0.016***	0.0%	1.000
Lsac	DP	0.4	-21.0%	1.000	0.0%	1.000
Lsac	COMBINED	0.4	14.9%	0.016***	0.0%	1.000
Adult	IF	0.0	4.3%	0.016***	1.0%	0.156
Adult	IF	0.1	6.1%	0.016***	19.4%	0.016***
Adult	IF	0.2	4.0%	0.016***	19.3%	0.016***
Adult	IF	0.3	-6.2%	0.891	22.8%	0.016***
Adult	IF	0.4	7.8%	0.500	28.3%	0.016***
Credit	IF	0.0	6.3%	0.016***	0.8%	0.156
Credit	IF	0.1	12.2%	0.109	12.3%	0.016***
Credit	IF	0.2	16.5%	0.031***	16.8%	0.016***
Credit	IF	0.3	21.8%	0.016***	16.6%	0.016***
Credit	IF	0.4	27.3%	0.047***	15.3%	0.016***
Lsac	IF	0.0	-26.4%	1.000	-45.9%	1.000
Lsac	IF	0.1	-14.6%	1.000	-5.1%	0.781
Lsac	IF	0.2	-44.1%	1.000	-4.0%	0.969
Lsac	IF	0.3	-10.2%	1.000	8.8%	0.016***
Lsac	IF	0.4	0.5%	0.500	11.3%	0.016***

Table 5: UTKFace REAL pilot (clean-test DP; n=6). Wins = seeds where DRO DP < Naive DP. Only α=0.4 DP/Combined reach p<0.05. Source: `results/utkface_summary.md`.

Attack	α	DP Naive	DP DRO	Wins	p
DP	0.1	0.048	0.051	1/6	0.969
DP	0.2	0.157	0.159	2/6	0.844
DP	0.4	0.222	0.213	6/6	0.016
Combined	0.1	0.031	0.033	2/6	0.953
Combined	0.4	0.144	0.136	6/6	0.016
IF	0.2	0.018	0.019	2/6	0.719
IF	0.4	0.016	0.016	4/6	0.422